Gas Prices, Inflation Expectations and the German Yield Curve

Evidence from the 2026 Iran War

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Data to 9 October 2026

Abstract

I ask whether European natural gas prices move the German government bond curve, and whether they do so through inflation expectations and the expected path of ECB policy. Using daily data from October 2014 to October 2026, I regress changes in inflation swaps, €STR overnight index swaps and Bund yields on returns of the Dutch TTF gas contract, controlling for Brent and the euro, and I let the relationship differ across four regimes. It does differ, sharply. Before 2021 gas barely moved Bund yields, and during the 2021–22 gas crisis higher gas prices came with *lower* yields. Since the Iran war began in March 2026, a 10% rise in TTF has lifted the one-year inflation swap by about 7bp, the expected policy rate one year ahead by 2 to 5bp and the two-year Bund by about 4bp on the same day, while the 30-year yield has not moved: a bear flattening. A break test with an unknown date puts the change in the first week of the war. Gas explains same-day curve moves, but it does not forecast later ones out of sample. Its forecasting content is for realised inflation 6 to 12 months ahead.

Keywords: natural gas, TTF, inflation swaps, ECB, Bund yield curve, regime dependence
JEL: E31, E43, E52, G12, Q43

1 Introduction

On the first trading day after Iran closed the Strait of Hormuz, Dutch TTF gas jumped by about 40% and the German two-year yield rose 16bp over two days. The 30-year rose 6bp. That pattern, a large move at the short end and a small one at the long end, repeated itself through the spring and summer of 2026 as the ECB raised its deposit rate in June and again in September. Between 27 February and 9 October the two-year Bund went from 2.03% to 3.09% and the 30-year from 3.36% to 3.85%.

The story most market participants told was simple: gas drives inflation, inflation expectations drive rate-hike expectations, and hike expectations drive the front end of the curve. This paper tests that chain link by link. I care about two separate claims. The first is transmission: do gas moves show up, on the day, in inflation swaps, €STR OIS and Bunds, after accounting for oil and the exchange rate? The second is predictability: does today's gas move tell us anything about curve moves in the days and weeks that follow?

The short answers are yes and no. Transmission is strong in the war regime and absent or reversed before it. In the war, a 10% gas move lifts the one-year inflation swap by 6.9bp ($p < 0.001$), the two-year Bund by 4.1bp ($p = 0.003$) and leaves the 30-year unchanged (0.2bp, $p = 0.886$), so the 2s30s slope falls by about 3.9bp. In 2021–22 the same regression gives −1.2bp for the two-year Bund. Predictability, on the other hand, is not there for the curve: local projections show no drift after the first day or two, and a gas-based forecast of next week's Bund move does no better than assuming no change. Where gas does help is in forecasting realised inflation, which responds with a lag of 6 to 12 months.

I think the regime result is the most useful one. It says the sign of the gas–rates relationship is set by how the market expects the central bank to react, not by gas itself. When the ECB was seen as patient and gas as a threat to growth, gas pushed yields down. Once the ECB had shown it would hike into an energy shock, the same gas move pushed the front end up.

The rest of the paper is organised as follows. Section 2 lays out the mechanism. Section 3 describes the data and the cleaning it needed, including a timing problem in the Bundesbank series that would otherwise produce spurious predictability. Section 4 sets out the regressions. Section 5 reports the results, Section 6 the diagnostics, and Section 7 discusses what they mean and where they stop. The appendix reports every OLS regression in full.

2 Mechanism

Figure 1 shows the chain. Gas sets the marginal price of electricity in much of Europe and feeds into household gas bills, fertiliser and industrial costs. Retail tariffs are regulated or contracted and reset slowly, so the effect on consumer prices arrives months after the wholesale move. Markets do not wait for those prints. Inflation swaps price expected HICP inflation directly: the one-year swap covers the next twelve months, and the 1y1y forward covers the year after that, which is closer to the horizon the ECB targets. If the ECB is expected to respond to higher expected inflation, €STR OIS rates rise, and with them the two-year Bund, which mostly tracks the expected policy path. Long yields depend more on long-run growth, long-run inflation and term premia, so a policy shock moves the short end more than the long end.

Two forces work against this chain. A gas supply shock is also a growth shock, and weaker growth pushes yields down, as does demand for safe assets in a crisis. Which force dominates

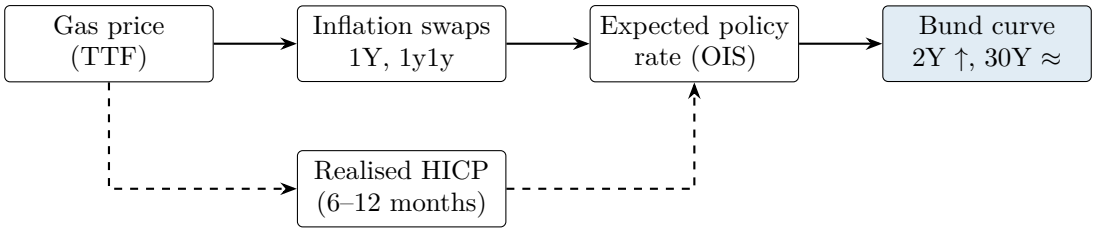


Figure 1: The transmission chain tested in this paper. Solid arrows: market prices that react on the day. Dashed: the slower path through realised inflation, which the ECB reacts to at later meetings.

depends on how the central bank is expected to respond. The second problem is oil. The Hormuz closure hit oil and gas at once, and euro inflation swaps have historically followed Brent more closely than gas, so any estimate of the gas effect has to control for oil. Adolfsen et al. (2026) add a further caution: in their BVAR, LNG supply shocks, which is what this war mostly is, move euro-area inflation expectations less than pipeline supply shocks do.

3 Data

Table 1 lists the series. The sample runs from October 2014 to 9 October 2026 at daily frequency. €STR OIS starts in October 2019 and the one-year inflation swap in April 2022, so regressions involving them use shorter samples. Bund yields come from the Bundesbank and are yields on the current (on-the-run) bond in each maturity bucket. Everything else comes from LSEG.

Table 1: Data

Series	Source / code	Definition	Start
TTF gas	LSEG TFMBMc1	ICE Endex Dutch TTF front-month futures, EUR/MWh	2014
Brent	LSEG LCOc1	ICE Brent front-month futures, USD/bbl	2014
EUR/USD, EUR/GBP	LSEG EUR=, EURGBP=	Spot mid	2014
Inflation swap 1Y	LSEG EUCPIZ1Y=TWEB	Zero-coupon swap on euro-area HICP ex tobacco, 1 year	2022
Inflation swap 1y1y	LSEG EUIL1YF1Y=R	One-year rate starting in one year	2016
€STR OIS	LSEG EUREST1M/3M/1Y/2Y=	Overnight index swaps; 1y1y forward derived from 1Y and 2Y	2019
Bund yields	Bundesbank BBSSY	Yields of current federal securities, 2Y to 30Y	1993–2026
5Y swap spread	LSEG DE5IRS5Y=RR	5Y swap rate minus 5Y Bund, bp	2014
HICP	Eurostat	Euro-area index, 2025 = 100, not seasonally adjusted	1999

Four data problems mattered. The file delivered as the "1Y inflation swap" is in fact the 1y1y forward; I relabelled it and added the true one-year swap. The HICP file splices two index bases, which creates a fake 34% drop in December 1999; I keep only the data on the current base. Front-month futures series jump when the contract rolls, so I identify roll days from jumps in open interest (about twelve a year) and drop those returns.

The fourth problem is timing, and it is the one that would have done the most damage. The Bundesbank observes its yields roughly half a trading day before the LSEG closes used for gas, swaps and OIS. Regressing the daily change in the two-year Bund on the same-day and previous-day change in the two-year OIS gives coefficients of 0.49 and 0.53, and the correlation between the two series rises from 0.44 at one day to 0.67 at two days and 0.87 at five. Ignored, this

mismatch makes yesterday's swap and gas moves look like they predict today's Bund move. I deal with it by adding the previous day's market moves to every daily regression with a Bundesbank series on the left-hand side, and by repeating the analysis at weekly frequency.

I split the sample into four regimes: 2014–21 (before the gas crisis, with the ECB at or below zero), July 2021 to December 2022 (the gas crisis), January 2023 to 27 February 2026, and the Iran war from 2 March 2026. Table 2 describes the daily changes. All series have fat tails (the excess kurtosis of the two-year Bund change is 15 in the full sample) and volatility clusters, which is why every standard error in this paper is heteroskedasticity- and autocorrelation-robust.

Table 2: Daily changes: full sample and Iran-war regime

Daily change	N	s.d.	Excess kurt.	AC(1)	N (war)	Mean (war)	s.d. (war)
TTF return (							
Brent return (							
ILS 1y1y (bp)	2733	3.73	9.2	0.07	157	0.30	4.50
OIS 1y1y (bp)	1784	6.08	4.4	0.02	157	0.82	7.65
Bund 2Y (bp)	3041	4.12	15.3	−0.03	157	0.68	5.35
Bund 10Y (bp)	3041	4.52	4.5	−0.01	157	0.50	4.32
Bund 30Y (bp)	3041	4.61	3.4	−0.00	157	0.31	3.41
2s10s (bp)	3041	3.01	4.4	−0.03	157	−0.18	2.82

Notes: Full sample October 2014 to 9 October 2026 (shorter for OIS and swaps). The war regime starts on 2 March 2026. TTF and Brent returns exclude contract-roll days.

4 Empirical strategy

4.1 Transmission regressions

Unit-root tests (Appendix Table 25) show that all the series are non-stationary in levels and stationary in first differences, and there is no robust cointegration between gas and yields (Table 26). I therefore work with changes. Let Δy_t be the daily change in basis points of an inflation swap, OIS rate, Bund yield or slope, and let g_t, o_t and e_t be 100 times the log change in TTF, Brent and EUR/USD. For series observed at the LSEG close the regression is

$$\Delta y_t = \alpha + \beta\, g_t + \gamma\, o_t + \delta\, e_t + \varepsilon_t, \tag{1}$$

and for Bundesbank yields and slopes, which are observed earlier in the day,

$$\Delta y_t = \alpha + \sum_{j=0}^{1} (\beta_j\, g_{t-j} + \gamma_j\, o_{t-j} + \delta_j\, e_{t-j}) + \varepsilon_t, \tag{2}$$

in the spirit of Dimson (1979). The total gas effect is β in (1) and $\beta_0 + \beta_1$ in (2). Because g_t is a log return in percent, the effect of a 10% rise in TTF is $\beta \times 100 \ln 1.1 \approx 9.53\,\beta$ basis points; this is the number I report in the text. Brent and EUR/USD enter as controls, not as objects of interest.

All regressions are estimated by OLS separately in each regime. Standard errors are Newey–West with a Bartlett kernel and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags (Newey and West, 1987), which gives 4 lags in the war regime ($T = 135$ for the Bund regressions) and 8 in the full sample ($T = 2642$). The weekly version uses Friday-to-Friday changes with 3 lags in the war regime ($T = 32$). Weekly gas and Brent returns are sums of the roll-cleaned daily returns, and for Bund series the weekly

market window ends on Thursday, so it closes before the Bundesbank's Friday observation.

To test whether the war changed the relationship I pool the full sample and interact every regressor with regime dummies,

$$\Delta y_t = \alpha + \sum_r \alpha_r D_t^r + \mathbf{x}_t'\boldsymbol{\theta} + \sum_{r \in \{\text{crisis, war}\}} D_t^r \, \mathbf{x}_t'\boldsymbol{\theta}_r + \varepsilon_t, \tag{3}$$

where $\mathbf{x}_t$ holds the regressors of (1) or (2), and I test that the gas elements of $\boldsymbol{\theta}_r$ sum to zero. I also run a Chow test at 2 March 2026 and a Andrews (1993) sup-Wald test that searches for the break date over the middle 70% of a 2023–26 sample. Because I test 20 dependent variables, I report Holm-adjusted p-values (Holm, 1979) as well as raw ones.

4.2 Channel, dynamics and forecasting

To ask how much of the gas effect runs through inflation expectations, I estimate a serial mediation model

$$M_{1t} = a_1 g_t + \ldots, \qquad M_{2t} = a_2 g_t + d_{21} M_{1t} + \ldots, \qquad \Delta y_t = c' g_t + b_1 M_{1t} + b_2 M_{2t} + \ldots \tag{4}$$

with M_1 the inflation swap, M_2 the OIS 1y1y and the same controls throughout. The indirect effects are $a_1 b_1$, $a_2 b_2$ and $a_1 d_{21} b_2$. Confidence intervals come from a moving-block bootstrap. The daily version stops at OIS, because all three series share the LSEG close; the full chain to the Bund runs at weekly frequency.

For dynamics I use local projections (Jordà, 2005),

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \, g_t + \gamma_h \, o_t + \delta_h \, e_t + \sum_{k=1}^{2} (\phi_{hk} \, g_{t-k} + \psi_{hk} \, \Delta y_{t-k}) + u_{t+h}, \quad h = 0, \ldots, 20, \tag{5}$$

and a predictive variant whose left-hand side starts after the impact day. I complement these with a VAR and Granger tests, and with expanding-window out-of-sample forecasts of next week's change evaluated with the Clark and West (2007) test. Finally, for the first link of the chain I regress year-on-year HICP inflation on 12-month changes in TTF and Brent (in euros) lagged $L = 0, \ldots, 12$ months.

5 Results

5.1 Gas reaches inflation with a lag

Oil and gas reach consumer prices on different clocks (Figure 2, Table 3). A 10% rise in Brent adds 0.30pp to year-on-year HICP inflation in the same month ($p = 0.011$). The same rise in TTF adds nothing on impact (−0.01pp), but its effect builds steadily to 0.21pp after nine months ($p = 0.020$) and 0.24pp after a year ($p = 0.007$). In a distributed-lag model of monthly inflation the gas lags are jointly significant ($p < 0.001$), while the first six months on their own are not ($p = 0.068$). This is what one would expect from regulated retail tariffs, and it matters for the rest of the paper: when gas jumps, the inflation the ECB will see arrives at the horizon it targets.

Table 3: Energy prices and realised inflation: HICP y/y on lagged 12-month energy price changes

Lag (months)	TTF, pp	p	Brent, pp	p	R^2	N
0	−0.01	0.918	0.30	0.011	0.17	130
3	0.10	0.216	0.20	0.076	0.33	127
6	0.15	0.065	0.17	0.185	0.50	124
9	0.21	0.020	0.04	0.767	0.58	121
12	0.24	0.007	−0.09	0.471	0.56	118

Notes: Each row is a regression of euro-area HICP inflation (y/y, pp) on the 12-month log change in TTF and Brent (in euros), both lagged L months; coefficients scaled to a +10% change. Monthly data from December 2015; Newey–West standard errors (12 lags).

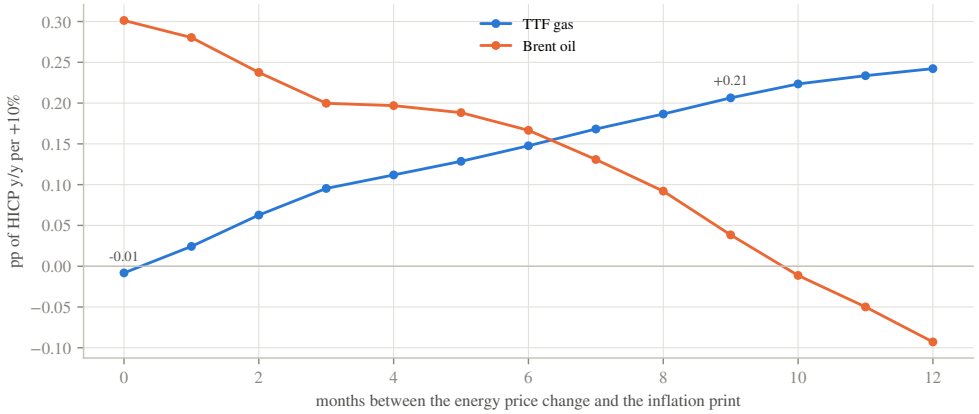


Figure 2: Effect of a 10% rise in energy prices on euro-area HICP inflation (y/y), by months of lag. Brent's effect is immediate and fades; TTF's builds over a year.

5.2 Same-day transmission in the war

Tables 4 and 5 report the war-regime regressions in full. Three things stand out. First, gas moves the front end. The two-year Bund rises 4.1bp per 10% gas move, almost all of it on the same day ($\hat{\beta}_0 = 0.404$ per 1% against $\hat{\beta}_1 = 0.024$). The effect falls with maturity: 2.0bp at five years, 1.3bp at ten and 0.2bp at thirty. Second, the fit is far better than in normal times. The two-year Bund regression has an R^2 of 0.40 (adjusted 0.37, robust $F = 20.9$), against 0.02 in the full sample. For the one-year inflation swap the R^2 is 0.67, and for the OIS 1y1y 0.58.

Third, oil is doing a lot of the work in the swap markets. Brent's effect on the one-year inflation swap is 17.4bp per 10%, against 6.9bp for gas, and on the OIS 1y1y it is 11.4bp against 2.3bp. For the Bund, Brent's effect is split evenly between the same day ($\hat{\gamma}_0 = 0.303$) and the previous day ($\hat{\gamma}_1 = 0.301$, $p = 0.039$). That fits the timing problem described above: Brent settles in the London evening, after the Bundesbank has taken its snapshot, so part of each day's oil move reaches the Bund series a day later. Gas keeps a significant effect on the two-year Bund after all of this. The raw p-value is $p = 0.003$; after Holm correction it is 0.036 within the 13 rate series and 0.057 across all 20 series tested, so it sits right at the conventional threshold under the strictest correction. The weekly estimate, which is free of the timing issue, is 3.6bp ($p = 0.001$; Appendix Table 21).

5.3 The relationship changed sign across regimes

Table 6 and Figure 3 put the four regimes side by side. Before 2021 gas had small, mostly insignificant effects, and where they were significant they steepened the curve: 1.0bp on 2s30s per

Table 4: OLS transmission regressions, Iran-war regime (daily, 2 March – 9 October 2026), part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.722^{***}	0.186^{**}	0.244^{**}	0.217^{**}
	(0.196)	(0.082)	(0.122)	(0.107)
o_t (Brent)	1.825^{***}	0.516^{***}	1.201^{***}	0.991^{***}
	(0.171)	(0.074)	(0.145)	(0.128)
e_t (EUR/USD)	1.484	−0.964	−2.105	−1.446
	(1.543)	(0.771)	(1.613)	(1.421)
Constant	0.898	−0.061	0.460	0.420
	(0.565)	(0.230)	(0.325)	(0.284)
TTF total, bp per +10%	6.88^{***}	1.77^{**}	2.32^{**}	2.07^{**}
	(1.87)	(0.78)	(1.16)	(1.02)
R^2	0.672	0.425	0.578	0.579
Adjusted R^2	0.665	0.412	0.569	0.570
Robust F (all slopes)	186.3^{***}	44.5^{***}	92.7^{***}	80.0^{***}
Residual s.e. (bp)	6.96	3.30	5.03	4.16
Durbin–Watson	2.17	2.17	2.34	2.31
HAC lags	4	4	4	4
Observations	142	143	143	143
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	0.404^{***}	0.248^{***}	0.183^{*}	0.040
	(0.105)	(0.089)	(0.094)	(0.087)
g_{t-1}	0.024	−0.037	−0.044	−0.020
	(0.095)	(0.083)	(0.086)	(0.071)
o_t (Brent)	0.303^{*}	0.333^{***}	0.269^{**}	0.201^{**}
	(0.164)	(0.125)	(0.118)	(0.088)
o_{t-1}	0.301^{**}	0.293^{***}	0.285^{***}	0.174^{**}
	(0.146)	(0.105)	(0.092)	(0.076)
e_t (EUR/USD)	1.200	1.107	1.356	1.135
	(0.834)	(0.810)	(0.839)	(0.777)
e_{t-1}	-1.862^{*}	-1.834^{**}	−1.206	−0.461
	(1.053)	(0.830)	(0.776)	(0.848)
Constant	0.530^{*}	0.467^{*}	0.434	0.274
	(0.313)	(0.276)	(0.270)	(0.263)
TTF total, bp per +10%	4.08^{***}	2.01	1.32	0.19
	(1.37)	(1.22)	(1.39)	(1.30)
R^2	0.397	0.377	0.302	0.146
Adjusted R^2	0.369	0.348	0.269	0.106
Robust F (all slopes)	20.9^{***}	21.5^{***}	13.6^{***}	3.5^{***}
Residual s.e. (bp)	4.05	3.49	3.34	2.92
Durbin–Watson	2.20	2.00	1.90	1.78
HAC lags	4	4	4	4
Observations	135	135	135	135

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: OLS transmission regressions, Iran-war regime (daily, 2 March – 9 October 2026), part 2: curve slopes

Panel C. Curve slopes	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	-0.156^{***}	-0.221^{***}	-0.364^{***}	-0.208^{***}
	(0.050)	(0.082)	(0.111)	(0.069)
g_{t-1}	−0.061	−0.068	−0.044	0.017
	(0.059)	(0.084)	(0.100)	(0.063)
o_t (Brent)	0.030	−0.034	−0.102	−0.132
	(0.063)	(0.095)	(0.132)	(0.081)
o_{t-1}	−0.008	−0.016	−0.128	−0.119
	(0.076)	(0.099)	(0.140)	(0.081)
e_t (EUR/USD)	−0.093	0.156	−0.065	0.028
	(0.335)	(0.583)	(0.722)	(0.514)
e_{t-1}	0.029	0.657	1.402	1.373
	(0.574)	(1.000)	(1.354)	(0.876)
Constant	−0.063	−0.096	−0.256	−0.193
	(0.128)	(0.210)	(0.309)	(0.207)
TTF total, bp per +10%	-2.07^{***}	-2.76^{**}	-3.89^{**}	-1.82^{*}
	(0.76)	(1.28)	(1.60)	(1.01)
R^2	0.106	0.152	0.253	0.295
Adjusted R^2	0.065	0.113	0.218	0.262
Robust F (all slopes)	2.0^{*}	3.3^{***}	6.5^{***}	9.8^{***}
Residual s.e. (bp)	1.88	2.74	3.75	2.48
Durbin–Watson	2.46	2.24	2.20	2.11
HAC lags	4	4	4	4
Observations	135	135	135	135

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

10% ($p = 0.002$). With the ECB at the lower bound the front end could not move, so whatever gas did showed up in long yields. During the 2021–22 crisis the sign flipped. Higher gas came with lower yields across the curve, −1.2bp for the two-year Bund ($p = 0.026$), and with lower expected policy rates (−4.0bp on the OIS 1y1y at weekly frequency, $p = 0.003$). Markets read gas as a recession risk and the ECB was still describing inflation as transitory. From 2023 to February 2026 the effects are close to zero everywhere. Then the war arrived.

The formal tests agree. In the pooled model (3) the war-regime gas effect on the two-year Bund exceeds the base-regime effect by 3.9bp ($p = 0.004$), and the difference is also significant for the one-year inflation swap ($p = 0.006$) and for 2s30s ($p = 0.018$). The sup-Wald test, which is not told when the war started, finds the largest break in the OIS 2Y gas sensitivity on 5 March 2026 (statistic 16.1 against a 5% critical value of 8.68) and in the OIS 1y1y on 11 March 2026 (14.6). For the two-year Bund the Wald test at the war date rejects ($p = 0.041$) but the unknown-date version does not reach significance (4.4), because the war sits close to the edge of the search window. The rolling estimates in Figure 4 tell the same story visually.

5.4 Which part of the curve moves

Every tenor rose during the war, and each rose less than the one before it: 106bp at two years, 95bp at five, 78bp at ten and 49bp at thirty (Figure 5). The regressions say gas contributed to

Table 6: Total TTF effect by regime, basis points per +10% gas move (daily)

	2014–21	2021–22 crisis	2023–Feb 26	Iran war	Full sample
ILS 1Y		6.49***	1.56***	6.88***	3.85***
		(2.22)	(0.47)	(1.87)	(0.83)
ILS 1y1y	0.35*	1.56***	1.06***	1.77**	1.29***
	(0.19)	(0.56)	(0.28)	(0.78)	(0.37)
OIS 1y1y	0.25	−1.04*	0.51	2.32**	0.15
	(0.18)	(0.58)	(0.62)	(1.16)	(0.39)
OIS 2Y	0.14	−0.73*	0.32	2.07**	0.14
	(0.12)	(0.40)	(0.47)	(1.02)	(0.28)
Bund 2Y	0.31	−1.24**	−0.34	4.08***	−0.19
	(0.20)	(0.56)	(1.08)	(1.37)	(0.39)
Bund 5Y	0.53*	−1.35**	−0.40	2.01	−0.18
	(0.28)	(0.67)	(1.03)	(1.22)	(0.42)
Bund 10Y	0.99**	−1.35**	−0.71	1.32	−0.20
	(0.39)	(0.64)	(0.90)	(1.39)	(0.40)
Bund 30Y	1.36***	−1.23*	−0.71	0.19	−0.22
	(0.45)	(0.70)	(0.76)	(1.30)	(0.44)
2s5s	0.22	−0.11	−0.07	−2.07***	0.01
	(0.18)	(0.30)	(0.26)	(0.76)	(0.15)
2s10s	0.68**	−0.11	−0.38	−2.76**	−0.01
	(0.28)	(0.39)	(0.42)	(1.28)	(0.21)
2s30s	1.05***	0.01	−0.38	−3.89**	−0.04
	(0.35)	(0.53)	(0.59)	(1.60)	(0.30)
5s30s	0.82***	0.12	−0.31	−1.82*	−0.04
	(0.28)	(0.37)	(0.44)	(1.01)	(0.22)

Notes: Each cell is a separate regression of the form in equation (2) (Bundesbank series) or (1) (LSEG series), estimated within the regime, controlling for Brent and EUR/USD. Newey–West standard errors in parentheses. The 1Y inflation swap starts in April 2022 and €STR OIS in October 2019. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

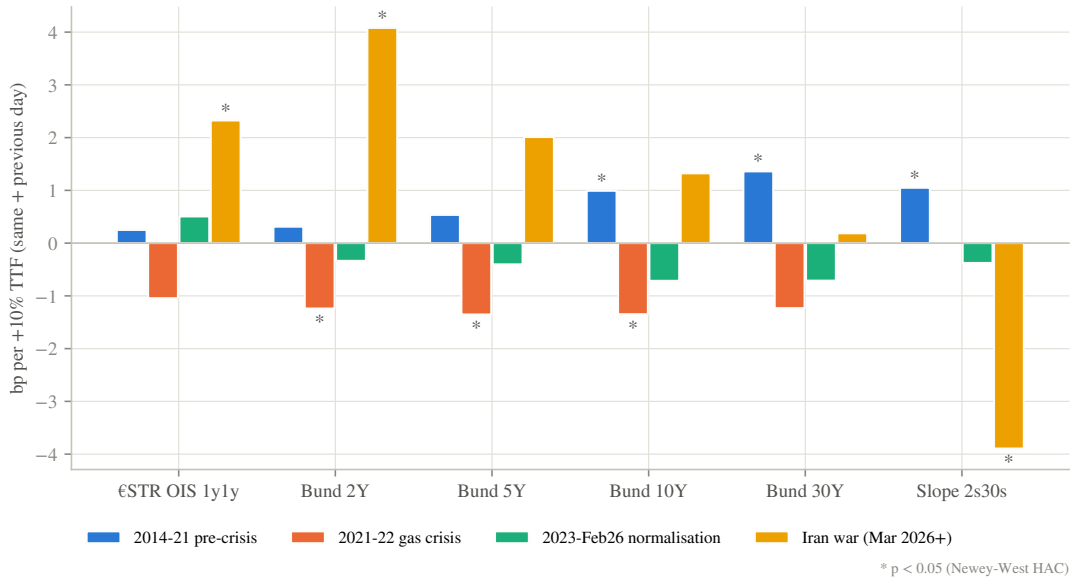


Figure 3: Total TTF effect by regime, bp per 10% gas move (daily). Asterisks mark $p < 0.05$. Only in the war does gas lift the front end and flatten the curve.

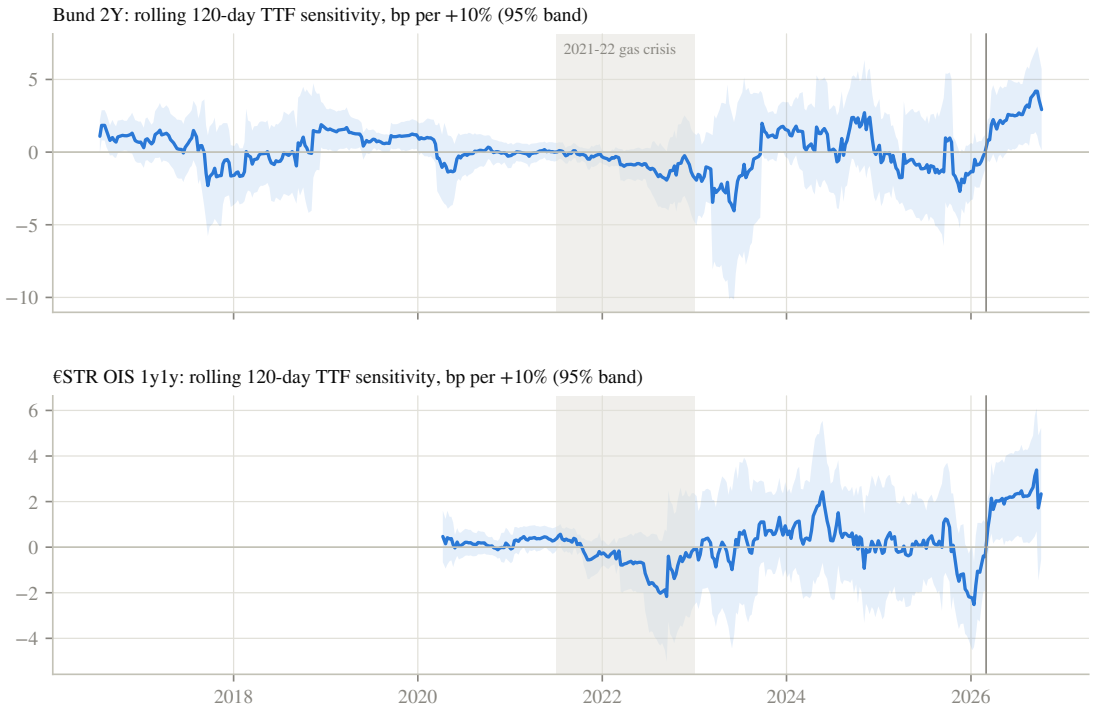


Figure 4: Rolling 120-day TTF sensitivity of the two-year Bund and the OIS 1y1y, bp per 10%, with 95% bands. Shaded: the 2021–22 gas crisis. Vertical line: 2 March 2026.

this shape. Table 7 compares six slopes. 2s30s carries the largest gas effect, −3.9bp per 10% daily and −3.0bp weekly, both significant, but Brent flattens it too. 2s10s (−2.8bp daily, $p = 0.031$) and 2s5s (−2.1bp, $p = 0.006$) respond to gas and not to oil, which makes them the cleaner gas measures; the 2s5s effect, though, is gone at weekly frequency (−0.4bp), as the five-year catches up with the two-year within the week. A principal-component decomposition of the curve (level 85% and slope 12% of daily variance) gives the same answer: in the war gas loads on the slope factor (−2.8bp, $p = 0.016$), less clearly on the level factor (3.6bp, $p = 0.118$).

Table 7: Which flattener carries gas? Slope sensitivities in the war regime

Slope	War, daily	p	War, weekly	p	Brent	p	War vs base p	27 Feb	9 Oct
2s5s	−2.07	0.006	−0.37	0.483	0.20	0.758	0.007	25	14
2s10s	−2.76	0.031	−1.52	0.092	−0.47	0.626	0.041	66	38
5s10s	−0.69	0.285	−1.15	0.008	−0.67	0.203	0.305	41	24
10s30s	−1.14	0.071	−1.49	< 0.001	−1.72	0.012	0.037	67	38
2s30s	−3.89	0.015	−3.01	0.009	−2.19	0.115	0.018	133	76
5s30s	−1.82	0.072	−2.64	< 0.001	−2.39	0.016	0.068	108	62

Notes: Slope = long minus short Bund yield, in bp; sensitivities in bp of slope change per +10% TTF (negative = flattening). Daily estimates include the previous day's market moves. "War vs base" tests whether the war-regime effect differs from 2014–21 and 2023–26 pooled. The last two columns give slope levels before the war and at the end of the sample.

5.5 How much runs through inflation expectations

The mediation estimates (Table 8) suggest the inflation-expectations channel is real but not the whole story. At daily frequency in the war, gas raises the OIS 1y1y by 2.3bp per 10%, and about 31% of that (0.7bp) runs through the 1y1y inflation swap: each basis point on the swap moves the OIS 1y1y by 0.41bp. The rest is a direct effect. My reading is that the market prices the ECB's reaction to energy news without waiting for inflation swaps to move first. At weekly

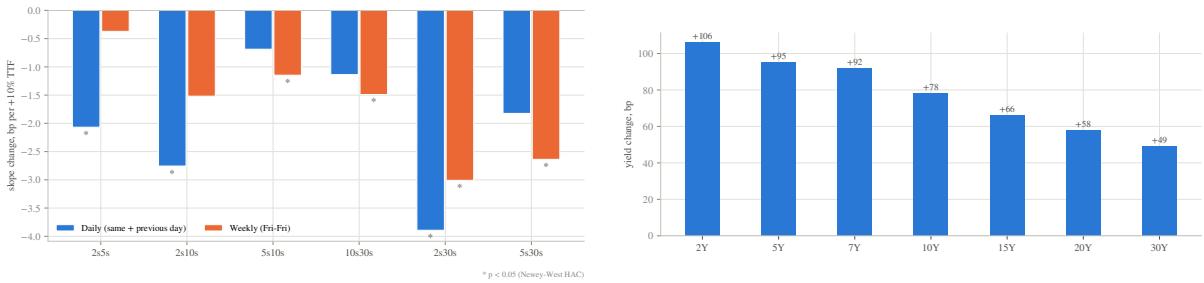


Figure 5: Left: slope change per 10% TTF in the war, daily and weekly (asterisks: $p < 0.05$). Right: change in Bund yields by tenor between 27 February and 9 October 2026.

frequency, the indirect path from gas through swaps and OIS to the two-year Bund totals 3.2bp and is significant, and each basis point of OIS 1y1y passes 0.71bp into the Bund (0.69 over the whole 2019–26 period).

Table 8: Mediation: how much of the gas effect runs through expectations? (Iran-war regime)

Effect	Estimate	95% CI low	95% CI high
Daily: TTF → ILS 1y1y → OIS 1y1y			
Total: TTF → OIS 1y1y	2.32	0.67	4.54
Direct: TTF → OIS 1y1y (not via ILS)	1.59	−0.24	3.92
Indirect: TTF → Infl. swap 1y1y → OIS 1y1y	0.73	0.09	2.06
path a: TTF → Infl. swap 1y1y (bp per 1%)	0.19	0.03	0.40
path b: Infl. swap 1y1y → OIS 1y1y (bp per bp)	0.41	0.16	0.71
Weekly: TTF → ILS / OIS → Bund 2Y			
Total: TTF → Bund 2Y	2.29	−2.45	6.18
Direct: TTF → Bund 2Y (not via ILS/OIS)	−0.91	−5.20	0.17
Indirect: TTF → ILS 1y1y → Bund 2Y	−0.18	−0.81	0.63
Indirect: TTF → OIS 1y1y → Bund 2Y	2.62	−0.54	7.73
Indirect: TTF → ILS 1y1y → OIS 1y1y → Bund 2Y	0.76	−0.40	1.83
Total indirect	3.20	0.19	8.20

Notes: Total, direct and indirect effects in bp per +10% TTF; path coefficients as labelled. Confidence intervals from a moving-block bootstrap (2,000 draws; blocks of 10 days or 4 weeks). Controls: Brent and EUR/USD.

5.6 Gas explains the curve but does not forecast it

The impact response is not followed by any drift (Figure 6). In the war, the cumulative response of the two-year Bund stays around its impact level for a week and then becomes too noisy to distinguish from zero; the predictive version of the local projection, which starts after the impact day, is insignificant at every horizon (at five days, $p = 0.478$). The 1y1y inflation swap even reverses: its cumulative response after 20 days is −8.5bp ($p = 0.004$), which is consistent with markets treating most of each gas spike as temporary.

The Granger tests agree. At weekly frequency TTF does not help predict the two-year Bund ($p = 0.706$), and in the war it does not predict OIS either ($p = 0.652$). Over 2019–26 there is a daily Granger effect from TTF to OIS ($p = 0.004$), but the local projection shows it is a small give-back of −0.8bp per 10% the next day, not momentum. Out of sample (Table 9) a gas model of next week's Bund change loses to the no-change forecast in the war (OOS R^2 of −0.5%). The only gain is for the OIS 1y1y in the war (5.2%, $p = 0.018$), from 31 forecasts, which I would not lean on.

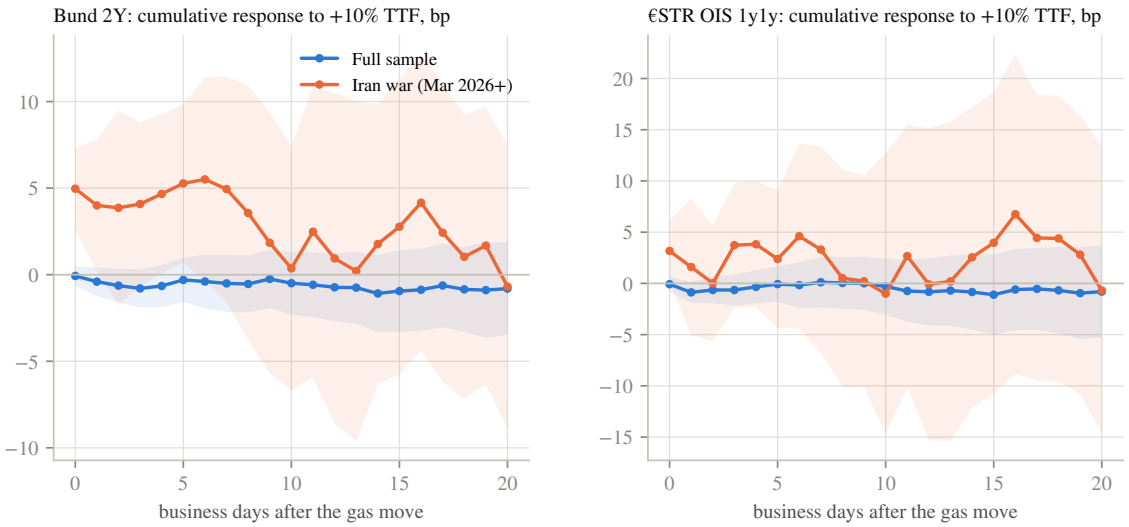


Figure 6: Cumulative response to a 10% TTF move over 20 business days (local projections, 95% bands): full sample and Iran-war regime.

Table 9: Out-of-sample forecasts of next week's change: gas model vs. no change

Target (next week)	Evaluation window	Forecasts	OOS R^2 (%)	Clark–West t	p
€STR OIS 2Y	Oct 2020 - Oct 2026	310	−0.6	0.47	0.318
€STR OIS 2Y	Iran war only	31	3.9	1.70	0.045
€STR OIS 1y1y	Oct 2020 - Oct 2026	309	−1.0	−0.04	0.516
€STR OIS 1y1y	Iran war only	31	5.2	2.09	0.018
Infl. swap 1y1y	Jan 2020 - Oct 2026	350	−1.2	−0.72	0.764
Infl. swap 1y1y	Iran war only	31	−4.0	−0.91	0.820
Bund 2Y	Jan 2020 - Oct 2026	350	−0.6	−0.26	0.602
Bund 2Y	Iran war only	31	−0.5	−0.15	0.559
Bund 10Y	Jan 2020 - Oct 2026	350	−0.8	−0.56	0.711
Bund 10Y	Iran war only	31	1.1	0.71	0.240
Slope 2s10s	Jan 2020 - Oct 2026	350	−1.0	−1.26	0.895
Slope 2s10s	Iran war only	31	−1.4	−1.73	0.958

Notes: Expanding-window OLS of next week's change on this week's TTF return. For Bund targets the TTF window ends on Thursday. OOS $R^2 = 1 - \text{MSE}_{model}/\text{MSE}_{benchmark}$. Clark–West (2007) one-sided test for nested models.

6 Diagnostics and robustness

Table 10 summarises the residual tests. Heteroskedasticity, ARCH effects and fat tails are present in almost every model, and serial correlation in several full-sample ones; the Newey–West errors are robust to all of them. CUSUM rejects stability for the full-sample Bund regressions, which is what motivates estimating by regime in the first place. The RESET test rejects linearity for the two-year Bund and 2s10s in the war, a hint that large gas days move rates more than proportionally. Collinearity between gas and oil is moderate: their correlation is 0.69 in the war, but no variance inflation factor exceeds 2.3.

The main results hold up at weekly frequency (Appendix Tables 21 and 23), where the Thursday-ending market window removes the timing problem for Bunds, and the two-year Bund and one-year inflation swap results survive the Holm correction discussed above. What I cannot rule out is that war headlines move gas, oil and rates at the same time for reasons that have nothing to do with inflation expectations; the Brent control absorbs some of that, not all of it.

Table 10: Residual diagnostics (p-values unless stated)

Model	Sample	DW	BG(5)	White	ARCH(5)	JB	RESET	CUSUM	VIF
Infl. swap 1y1y	Full sample	1.83	< 0.001	< 0.001	< 0.001	< 0.001	0.105	0.628	1.06
Infl. swap 1y1y	Iran war (Mar 2026+)	2.17	0.070	< 0.001	< 0.001	0.032	0.159	0.300	1.84
€STR OIS 1y1y	Full sample	2.01	0.722	< 0.001	< 0.001	< 0.001	< 0.001	0.298	1.06
€STR OIS 1y1y	Iran war (Mar 2026+)	2.34	< 0.001	< 0.001	0.009	0.008	0.555	0.996	1.84
Bund 2Y	Full sample	2.10	< 0.001	< 0.001	< 0.001	< 0.001	0.011	0.025	1.05
Bund 2Y	Iran war (Mar 2026+)	2.20	0.277	< 0.001	< 0.001	< 0.001	0.034	0.919	2.35
Bund 30Y	Full sample	2.02	0.277	< 0.001	< 0.001	< 0.001	0.001	0.016	1.05
Bund 30Y	Iran war (Mar 2026+)	1.78	0.263	0.641	0.822	0.877	0.445	0.989	2.35
Slope 2s10s	Full sample	2.05	0.650	< 0.001	< 0.001	< 0.001	< 0.001	0.719	1.05
Slope 2s10s	Iran war (Mar 2026+)	2.24	0.277	< 0.001	0.037	0.016	0.016	0.726	2.35

Notes: BG: Breusch–Godfrey test for serial correlation up to lag 5; White: heteroskedasticity; ARCH: Engle's LM test; JB: Jarque–Bera normality; RESET: Ramsey test (fitted powers up to 3); CUSUM: stability of OLS residuals; VIF: variance inflation factor of the TTF regressor.

7 Discussion and limits

Three conclusions seem solid to me. The first is that the gas–rates link is conditional. Its sign depends on whether the market expects the central bank to treat an energy shock as an inflation problem or a growth problem. In 2022 it was the latter; in 2026, after two hikes, it is the former. Anyone using gas as a rates signal should watch the rolling sensitivity rather than assume a fixed beta. The second is that gas is priced as a shock to the expected policy path, not to long-run inflation. The one-year swap moves four times as much as in normal times, the 1y1y much less, and the 30-year yield not at all. That is the signature of a bear flattening. The third is that the predictive value of gas lies in the inflation data it foreshadows, not in tradable lead–lag relationships with the curve.

The limits are just as clear. The war regime is short, 143 trading days and 32 weeks, which is enough to reject a zero effect but not to pin down magnitudes. Oil and gas moved together in this episode, and a VAR can only bound gas's share of the variance of the expected policy rate: between 2% and 40% depending on the ordering. Identification rests on controls rather than on an external instrument; LNG-specific outage news or intraday windows around gas moves would sharpen it. The RESET results invite a threshold model. And the analysis would benefit from the HICP energy sub-index, the 5y5y inflation swap, a volatility index for risk-off days, and a constant-maturity Bund curve to remove on-the-run effects.

8 Conclusion

Since March 2026 European gas prices have been a significant driver of euro inflation swaps, of the expected ECB path and of the front end of the German curve, and they have flattened it. This was not true before, and in 2021–22 the opposite was true. Gas explains curve moves on the day they happen but does not forecast them; what it does forecast is realised inflation a year out. The practical lesson is to treat gas as a regime-dependent factor for the front end and the slope of the curve, and to check whether the regime still holds before relying on it.

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A Full OLS results

This appendix reports every transmission regression. Daily tables follow equations (1) and (2); weekly tables use Friday-to-Friday changes. Coefficients on g, o and e are in basis points per 1% move. The "TTF total" row converts the gas coefficients into the effect of a 10% move.

Table 11: OLS transmission regressions, 2014–21 regime (daily), part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)		0.036^{*}	0.026	0.015
		(0.019)	(0.019)	(0.013)
o_t (Brent)		0.357^{***}	0.030	0.034
		(0.051)	(0.049)	(0.040)
e_t (EUR/USD)		0.451^{**}	−0.537	−0.398
		(0.209)	(0.381)	(0.314)
Constant		0.015	0.080	0.063
		(0.061)	(0.080)	(0.058)
TTF total, bp per +10%		0.35^{*}	0.25	0.14
		(0.19)	(0.18)	(0.12)
R^2		0.195	0.032	0.041
Adjusted R^2		0.193	0.025	0.033
Robust F (all slopes)		24.9^{***}	1.5	1.5
Residual s.e. (bp)		2.04	1.59	1.14
Durbin–Watson		1.92	1.99	2.02
HAC lags		7	5	5
Observations		1271	403	404
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	0.006	0.021	0.057^{*}	0.058
	(0.015)	(0.024)	(0.030)	(0.037)
g_{t-1}	0.027^{*}	0.035	0.047	0.084^{**}
	(0.016)	(0.022)	(0.029)	(0.036)
o_t (Brent)	0.040	0.069^{*}	0.097^{*}	0.155^{**}
	(0.032)	(0.042)	(0.057)	(0.076)
o_{t-1}	0.026	0.016	0.075	0.102
	(0.022)	(0.036)	(0.052)	(0.065)
e_t (EUR/USD)	0.302^{**}	0.430^{**}	0.498^{*}	0.478
	(0.135)	(0.195)	(0.267)	(0.320)
e_{t-1}	0.352^{*}	0.583^{**}	0.596^{*}	0.628^{*}
	(0.180)	(0.241)	(0.320)	(0.360)
Constant	−0.011	−0.048	−0.073	−0.135
	(0.046)	(0.070)	(0.089)	(0.105)
TTF total, bp per +10%	0.31	0.53^{*}	0.99^{**}	1.36^{***}
	(0.20)	(0.28)	(0.39)	(0.45)
R^2	0.024	0.025	0.025	0.028
Adjusted R^2	0.020	0.021	0.021	0.024
Robust F (all slopes)	1.8^{*}	2.4^{**}	2.1^{*}	2.9^{***}
Residual s.e. (bp)	1.81	2.70	3.58	4.30
Durbin–Watson	2.15	2.03	2.02	2.04
HAC lags	7	7	7	7
Observations	1473	1473	1473	1473

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12: OLS transmission regressions, 2014–21 regime (daily), part 2: curve slopes

Panel C. Curve slopes	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	0.015	0.051^{**}	0.052^{*}	0.038
	(0.016)	(0.022)	(0.031)	(0.024)
g_{t-1}	0.009	0.020	0.058^{**}	0.049^{**}
	(0.014)	(0.021)	(0.029)	(0.025)
o_t (Brent)	0.030	0.057^{*}	0.115^{**}	0.085^{*}
	(0.021)	(0.034)	(0.052)	(0.049)
o_{t-1}	−0.009	0.050	0.077	0.086^{**}
	(0.021)	(0.039)	(0.057)	(0.043)
e_t (EUR/USD)	0.128	0.196	0.176	0.048
	(0.105)	(0.182)	(0.250)	(0.205)
e_{t-1}	0.231^{*}	0.244	0.276	0.045
	(0.125)	(0.254)	(0.339)	(0.282)
Constant	−0.036	−0.062	−0.124	−0.088
	(0.042)	(0.066)	(0.085)	(0.069)
TTF total, bp per +10%	0.22	0.68^{**}	1.05^{***}	0.82^{***}
	(0.18)	(0.28)	(0.35)	(0.28)
R^2	0.009	0.013	0.017	0.016
Adjusted R^2	0.005	0.009	0.013	0.012
Robust F (all slopes)	1.9^{*}	1.9^{*}	3.1^{***}	3.2^{***}
Residual s.e. (bp)	1.76	2.79	3.74	3.06
Durbin–Watson	2.25	2.09	2.12	2.14
HAC lags	7	7	7	7
Observations	1473	1473	1473	1473

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 13: OLS transmission regressions, 2021–22 gas-crisis regime (daily), part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.681^{***}	0.164^{***}	-0.110^{*}	-0.077^{*}
	(0.233)	(0.059)	(0.061)	(0.042)
o_t (Brent)	1.824^{***}	0.897^{***}	0.442^{**}	0.331^{**}
	(0.573)	(0.161)	(0.210)	(0.156)
e_t (EUR/USD)	-2.831^{*}	-0.694	-1.552	-0.971
	(1.511)	(0.545)	(1.388)	(0.992)
Constant	1.956	0.332	1.261^{***}	1.179^{***}
	(1.233)	(0.331)	(0.431)	(0.320)
TTF total, bp per +10%	6.49^{***}	1.56^{***}	-1.04^{*}	-0.73^{*}
	(2.22)	(0.56)	(0.58)	(0.40)
R^2	0.202	0.227	0.030	0.027
Adjusted R^2	0.187	0.220	0.022	0.019
Robust F (all slopes)	3.8^{**}	13.8^{***}	2.9^{**}	2.9^{**}
Residual s.e. (bp)	13.08	5.41	8.12	6.05
Durbin–Watson	1.59	1.60	1.93	1.99
HAC lags	4	5	5	5
Observations	169	352	352	352
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	-0.122^{**}	-0.120^{**}	-0.113^{**}	-0.087
	(0.051)	(0.059)	(0.056)	(0.057)
g_{t-1}	-0.008	-0.022	-0.028	-0.042
	(0.042)	(0.048)	(0.043)	(0.044)
o_t (Brent)	0.214	0.281	0.133	0.070
	(0.167)	(0.180)	(0.155)	(0.158)
o_{t-1}	0.248^{**}	0.315^{***}	0.429^{***}	0.404^{***}
	(0.117)	(0.121)	(0.115)	(0.109)
e_t (EUR/USD)	-0.020	-0.100	-0.418	0.107
	(0.677)	(0.723)	(0.676)	(0.716)
e_{t-1}	-0.284	-1.258	-1.227	-0.349
	(0.886)	(0.908)	(0.748)	(0.642)
Constant	1.231^{***}	1.221^{***}	1.014^{***}	0.777^{**}
	(0.339)	(0.354)	(0.336)	(0.335)
TTF total, bp per +10%	-1.24^{**}	-1.35^{**}	-1.35^{**}	-1.23^{*}
	(0.56)	(0.67)	(0.64)	(0.70)
R^2	0.036	0.046	0.063	0.052
Adjusted R^2	0.018	0.029	0.046	0.035
Robust F (all slopes)	2.7^{**}	3.1^{***}	5.1^{***}	3.7^{***}
Residual s.e. (bp)	6.50	6.87	5.97	5.54
Durbin–Watson	2.00	2.11	2.04	1.98
HAC lags	5	5	5	5
Observations	335	335	335	335

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14: OLS transmission regressions, 2021–22 gas-crisis regime (daily), part 2: curve slopes

Panel C. Curve slopes	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	0.003	0.009	0.035	0.033
	(0.028)	(0.036)	(0.050)	(0.035)
g_{t-1}	−0.015	−0.021	−0.035	−0.020
	(0.022)	(0.025)	(0.039)	(0.032)
o_t (Brent)	0.067	−0.081	−0.144	-0.211^{*}
	(0.074)	(0.102)	(0.140)	(0.110)
o_{t-1}	0.067	0.180^{***}	0.156	0.089
	(0.058)	(0.066)	(0.101)	(0.088)
e_t (EUR/USD)	−0.081	−0.398	0.127	0.208
	(0.396)	(0.566)	(0.762)	(0.565)
e_{t-1}	-0.973^{***}	-0.943^{**}	−0.064	0.909
	(0.296)	(0.409)	(0.680)	(0.597)
Constant	−0.009	−0.216	−0.454	-0.445^{**}
	(0.148)	(0.214)	(0.285)	(0.217)
TTF total, bp per +10%	−0.11	−0.11	0.01	0.12
	(0.30)	(0.39)	(0.53)	(0.37)
R^2	0.031	0.034	0.011	0.027
Adjusted R^2	0.014	0.017	-0.007	0.009
Robust F (all slopes)	2.8^{**}	3.0^{***}	0.7	1.3
Residual s.e. (bp)	3.19	3.96	5.82	4.94
Durbin–Watson	2.36	1.96	2.06	2.28
HAC lags	5	5	5	5
Observations	335	335	335	335

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: OLS transmission regressions, January 2023 – February 2026 regime (daily), part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.163^{***}	0.112^{***}	0.053	0.034
	(0.050)	(0.029)	(0.065)	(0.049)
o_t (Brent)	1.310^{***}	0.760^{***}	0.574^{***}	0.469^{***}
	(0.154)	(0.090)	(0.150)	(0.130)
e_t (EUR/USD)	0.074	−0.244	0.031	0.209
	(0.409)	(0.348)	(0.721)	(0.603)
Constant	−0.280	−0.035	−0.023	−0.064
	(0.206)	(0.106)	(0.207)	(0.165)
TTF total, bp per +10%	1.56^{***}	1.06^{***}	0.51	0.32
	(0.47)	(0.28)	(0.62)	(0.47)
R^2	0.223	0.215	0.035	0.036
Adjusted R^2	0.220	0.212	0.031	0.032
Robust F (all slopes)	35.0^{***}	38.7^{***}	5.7^{***}	5.1^{***}
Residual s.e. (bp)	4.86	2.98	5.84	4.62
Durbin–Watson	1.82	2.06	2.09	2.15
HAC lags	6	6	6	6
Observations	735	735	735	735
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	0.062	0.047	0.022	−0.003
	(0.065)	(0.067)	(0.062)	(0.051)
g_{t-1}	−0.097	−0.089	−0.097	−0.071
	(0.083)	(0.072)	(0.063)	(0.054)
o_t (Brent)	0.256^{**}	0.318^{**}	0.305^{***}	0.316^{***}
	(0.130)	(0.133)	(0.116)	(0.109)
o_{t-1}	0.519^{***}	0.508^{***}	0.487^{***}	0.427^{***}
	(0.139)	(0.126)	(0.111)	(0.099)
e_t (EUR/USD)	0.392	0.841	0.704	0.729
	(0.674)	(0.628)	(0.569)	(0.509)
e_{t-1}	−0.757	-1.464^{**}	-1.674^{***}	-1.695^{***}
	(0.557)	(0.638)	(0.640)	(0.544)
Constant	−0.069	0.011	0.040	0.130
	(0.180)	(0.174)	(0.163)	(0.157)
TTF total, bp per +10%	−0.34	−0.40	−0.71	−0.71
	(1.08)	(1.03)	(0.90)	(0.76)
R^2	0.047	0.061	0.069	0.074
Adjusted R^2	0.039	0.053	0.061	0.066
Robust F (all slopes)	3.3^{***}	4.5^{***}	6.0^{***}	6.7^{***}
Residual s.e. (bp)	5.42	5.45	5.02	4.58
Durbin–Watson	2.23	2.20	2.13	2.10
HAC lags	6	6	6	6
Observations	699	699	699	699

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16: OLS transmission regressions, January 2023 – February 2026 regime (daily), part 2: curve slopes

Panel C. Curve slopes	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	−0.014	−0.040	−0.065*	−0.050*
	(0.021)	(0.027)	(0.037)	(0.030)
g_{t-1}	0.007	−0.000	0.025	0.018
	(0.022)	(0.035)	(0.050)	(0.034)
o_t (Brent)	0.061	0.049	0.060	−0.002
	(0.041)	(0.065)	(0.097)	(0.081)
o_{t-1}	−0.011	−0.032	−0.093	−0.081
	(0.048)	(0.079)	(0.104)	(0.076)
e_t (EUR/USD)	0.449**	0.311	0.336	−0.112
	(0.215)	(0.326)	(0.476)	(0.371)
e_{t-1}	−0.707***	−0.917***	−0.939**	−0.232
	(0.257)	(0.340)	(0.390)	(0.289)
Constant	0.080	0.110	0.200	0.120
	(0.073)	(0.115)	(0.148)	(0.096)
TTF total, bp per +10%	−0.07	−0.38	−0.38	−0.31
	(0.26)	(0.42)	(0.59)	(0.44)
R^2	0.035	0.023	0.018	0.008
Adjusted R^2	0.027	0.014	0.010	-0.000
Robust F (all slopes)	2.9***	2.8**	2.5**	0.9
Residual s.e. (bp)	2.08	3.03	4.08	3.09
Durbin–Watson	2.09	2.06	2.11	2.30
HAC lags	6	6	6	6
Observations	699	699	699	699

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 17: OLS transmission regressions, full sample (daily), part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.404^{***}	0.135^{***}	0.016	0.015
	(0.087)	(0.038)	(0.040)	(0.030)
o_t (Brent)	1.611^{***}	0.516^{***}	0.456^{***}	0.375^{***}
	(0.122)	(0.049)	(0.100)	(0.080)
e_t (EUR/USD)	−0.831	−0.003	−0.680	−0.380
	(0.518)	(0.191)	(0.586)	(0.444)
Constant	0.289	0.054	0.356^{**}	0.306^{***}
	(0.286)	(0.064)	(0.148)	(0.116)
TTF total, bp per +10%	3.85^{***}	1.29^{***}	0.15	0.14
	(0.83)	(0.37)	(0.39)	(0.28)
R^2	0.291	0.199	0.049	0.053
Adjusted R^2	0.289	0.198	0.047	0.051
Robust F (all slopes)	61.6^{***}	44.5^{***}	8.4^{***}	8.4^{***}
Residual s.e. (bp)	7.33	3.14	5.89	4.57
Durbin–Watson	1.71	1.83	2.01	2.06
HAC lags	6	8	7	7
Observations	1046	2501	1633	1634
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	−0.001	−0.005	−0.004	−0.015
	(0.034)	(0.038)	(0.036)	(0.036)
g_{t-1}	−0.018	−0.014	−0.017	−0.009
	(0.029)	(0.028)	(0.026)	(0.025)
o_t (Brent)	0.166^{***}	0.192^{***}	0.174^{***}	0.185^{***}
	(0.045)	(0.046)	(0.044)	(0.052)
o_{t-1}	0.172^{***}	0.164^{***}	0.210^{***}	0.203^{***}
	(0.040)	(0.044)	(0.050)	(0.055)
e_t (EUR/USD)	0.275	0.421^{**}	0.383	0.461^{*}
	(0.201)	(0.212)	(0.233)	(0.253)
e_{t-1}	−0.074	−0.222	−0.231	−0.045
	(0.215)	(0.257)	(0.280)	(0.282)
Constant	0.172^{**}	0.167^{**}	0.126	0.063
	(0.078)	(0.085)	(0.087)	(0.089)
TTF total, bp per +10%	−0.19	−0.18	−0.20	−0.22
	(0.39)	(0.42)	(0.40)	(0.44)
R^2	0.023	0.024	0.026	0.025
Adjusted R^2	0.021	0.021	0.024	0.023
Robust F (all slopes)	5.2^{***}	4.7^{***}	5.2^{***}	5.1^{***}
Residual s.e. (bp)	4.07	4.39	4.42	4.54
Durbin–Watson	2.10	2.08	2.02	2.02
HAC lags	8	8	8	8
Observations	2642	2642	2642	2642

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 18: OLS transmission regressions, full sample (daily), part 2: curve slopes

Panel C. Curve slopes				
	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	−0.003	−0.003	−0.013	−0.010
	(0.014)	(0.018)	(0.026)	(0.019)
g_{t-1}	0.004	0.002	0.010	0.005
	(0.012)	(0.015)	(0.022)	(0.017)
o_t (Brent)	0.026	0.008	0.018	−0.008
	(0.018)	(0.030)	(0.049)	(0.044)
o_{t-1}	−0.008	0.038	0.031	0.039
	(0.019)	(0.031)	(0.043)	(0.031)
e_t (EUR/USD)	0.146	0.108	0.187	0.041
	(0.101)	(0.164)	(0.225)	(0.176)
e_{t-1}	−0.148	−0.157	0.029	0.177
	(0.114)	(0.192)	(0.256)	(0.211)
Constant	−0.005	−0.046	−0.109	−0.104*
	(0.036)	(0.056)	(0.073)	(0.055)
TTF total, bp per +10%	0.01	−0.01	−0.04	−0.04
	(0.15)	(0.21)	(0.30)	(0.22)
R^2	0.004	0.002	0.001	0.002
Adjusted R^2	0.001	-0.000	-0.001	-0.000
Robust F (all slopes)	1.0	0.6	0.5	0.5
Residual s.e. (bp)	2.10	3.05	4.19	3.37
Durbin–Watson	2.24	2.05	2.10	2.20
HAC lags	8	8	8	8
Observations	2642	2642	2642	2642

Notes: OLS on daily changes. Dependent variables in basis points; g and o are 100 times the log change of front-month TTF and Brent (roll days removed), e the same for EUR/USD, so coefficients are bp per 1% move. Bund yields and slopes are snapshotted by the Bundesbank about half a trading day before the LSEG closes, so their regressions add the previous day's market moves; the TTF total is $(\hat{\beta}_0 + \hat{\beta}_1) \times 100 \ln 1.1$. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 19: OLS transmission regressions, weekly, 2021–22 gas-crisis regime, part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	1.889^{***}	0.007	-0.423^{***}	-0.302^{***}
	(0.623)	(0.147)	(0.145)	(0.102)
o_t (Brent)	2.180^{**}	1.614^{***}	0.166	0.064
	(1.046)	(0.383)	(0.340)	(0.239)
e_t (EUR/USD)	−5.760	−2.180	−4.274	−3.104
	(5.492)	(1.596)	(2.901)	(1.931)
Constant	−2.310	0.298	4.434^{**}	4.507^{***}
	(7.777)	(1.377)	(1.979)	(1.476)
TTF total, bp per +10%	18.00^{***}	0.07	-4.03^{***}	-2.88^{***}
	(5.94)	(1.40)	(1.38)	(0.97)
R^2	0.299	0.262	0.132	0.134
Adjusted R^2	0.237	0.233	0.097	0.099
Robust F (all slopes)	3.7^{**}	8.9^{***}	3.5^{**}	3.2^{**}
Residual s.e. (bp)	50.37	14.34	18.25	13.04
Durbin–Watson	1.67	2.22	1.97	1.84
HAC lags	3	3	3	3
Observations	38	79	79	79
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	-0.183^{**}	-0.215^{**}	-0.201^{**}	-0.194^{**}
	(0.081)	(0.087)	(0.087)	(0.095)
o_t (Brent)	0.294	0.486	0.527^{*}	0.625^{**}
	(0.282)	(0.305)	(0.280)	(0.285)
e_t (EUR/USD)	0.687	−0.706	−1.670	−1.549
	(1.545)	(1.920)	(1.678)	(1.288)
Constant	4.398^{***}	3.797^{**}	3.107^{*}	2.387
	(1.574)	(1.682)	(1.617)	(1.629)
TTF total, bp per +10%	-1.74^{**}	-2.05^{**}	-1.92^{**}	-1.85^{**}
	(0.77)	(0.83)	(0.83)	(0.91)
R^2	0.065	0.072	0.090	0.107
Adjusted R^2	0.027	0.035	0.054	0.072
Robust F (all slopes)	2.3^{*}	3.5^{**}	4.7^{***}	3.5^{**}
Residual s.e. (bp)	14.05	14.59	12.87	12.18
Durbin–Watson	2.10	2.10	2.00	1.70
HAC lags	3	3	3	3
Observations	79	79	79	79

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 20: OLS transmission regressions, weekly, 2021–22 gas-crisis regime, part 2: curve slopes

Panel C. Curve slopes				
	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	-0.033	-0.019	-0.012	0.021
	(0.029)	(0.043)	(0.062)	(0.057)
o_t (Brent)	0.192^{**}	0.233^{*}	0.332	0.140
	(0.082)	(0.135)	(0.222)	(0.189)
e_t (EUR/USD)	-1.392^{**}	-2.357^{***}	-2.236^{***}	-0.843
	(0.601)	(0.608)	(0.800)	(0.905)
Constant	-0.601	-1.291^{*}	-2.012^{*}	-1.411^{*}
	(0.541)	(0.751)	(1.037)	(0.853)
TTF total, bp per +10%	-0.31	-0.18	-0.11	0.20
	(0.27)	(0.41)	(0.59)	(0.54)
R^2	0.133	0.134	0.074	0.020
Adjusted R^2	0.098	0.099	0.037	-0.019
Robust F (all slopes)	3.4^{**}	5.6^{***}	3.7^{**}	0.8
Residual s.e. (bp)	4.53	7.40	10.50	9.03
Durbin–Watson	2.06	2.20	2.23	2.32
HAC lags	3	3	3	3
Observations	79	79	79	79

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 21: OLS transmission regressions, weekly, Iran-war regime, part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.424	0.270***	0.501***	0.407***
	(0.312)	(0.082)	(0.172)	(0.154)
o_t (Brent)	2.912***	0.320*	0.545*	0.512**
	(0.513)	(0.176)	(0.326)	(0.255)
e_t (EUR/USD)	3.227	−1.041	−1.909	−0.943
	(3.479)	(1.707)	(1.751)	(1.566)
Constant	−0.239	−0.063	1.253	1.310
	(2.739)	(0.825)	(1.350)	(1.175)
TTF total, bp per +10%	4.04	2.57***	4.77***	3.88***
	(2.97)	(0.78)	(1.64)	(1.47)
R^2	0.695	0.467	0.537	0.543
Adjusted R^2	0.662	0.410	0.487	0.494
Robust F (all slopes)	67.3***	15.3***	56.8***	52.7***
Residual s.e. (bp)	19.78	6.68	10.40	8.67
Durbin–Watson	2.13	2.13	2.32	2.21
HAC lags	3	3	3	3
Observations	32	32	32	32
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	0.380***	0.341***	0.221***	0.065
	(0.117)	(0.083)	(0.083)	(0.085)
o_t (Brent)	0.547***	0.456***	0.432***	0.329***
	(0.164)	(0.140)	(0.144)	(0.122)
e_t (EUR/USD)	−0.987	−0.808	0.343	1.673
	(2.285)	(1.630)	(1.520)	(1.410)
Constant	1.097	1.052	1.106	1.054**
	(1.105)	(0.889)	(0.728)	(0.498)
TTF total, bp per +10%	3.62***	3.25***	2.11***	0.61
	(1.12)	(0.79)	(0.79)	(0.81)
R^2	0.484	0.513	0.442	0.299
Adjusted R^2	0.429	0.461	0.382	0.224
Robust F (all slopes)	18.8***	28.9***	15.8***	5.9***
Residual s.e. (bp)	8.83	7.19	6.47	5.30
Durbin–Watson	2.29	2.58	2.92	3.01
HAC lags	3	3	3	3
Observations	32	32	32	32

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 22: OLS transmission regressions, weekly, Iran-war regime, part 2: curve slopes

Panel C. Curve slopes	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	-0.039	-0.159^{*}	-0.316^{***}	-0.277^{***}
	(0.056)	(0.095)	(0.121)	(0.075)
o_t (Brent)	-0.092	-0.115^{**}	-0.218^{**}	-0.126
	(0.057)	(0.050)	(0.094)	(0.080)
e_t (EUR/USD)	0.179	1.330	2.661	2.481^{*}
	(1.094)	(1.765)	(2.516)	(1.473)
Constant	-0.045	0.009	-0.043	0.002
	(0.453)	(0.712)	(1.032)	(0.715)
TTF total, bp per +10%	-0.37	-1.52^{*}	-3.01^{***}	-2.64^{***}
	(0.53)	(0.90)	(1.16)	(0.71)
R^2	0.103	0.237	0.352	0.446
Adjusted R^2	0.007	0.155	0.282	0.387
Robust F (all slopes)	1.5	6.9^{***}	10.9^{***}	25.0^{***}
Residual s.e. (bp)	3.48	5.36	7.96	5.33
Durbin–Watson	2.04	1.89	1.81	1.67
HAC lags	3	3	3	3
Observations	32	32	32	32

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 23: OLS transmission regressions, weekly, full sample, part 1: expectations and yields

Panel A. Inflation and policy expectations				
	ILS 1Y	ILS 1y1y	OIS 1y1y	OIS 2Y
g_t (TTF)	0.934^{***}	0.068	-0.084	-0.055
	(0.331)	(0.082)	(0.110)	(0.080)
o_t (Brent)	1.899^{***}	0.650^{***}	0.460^{***}	0.382^{***}
	(0.300)	(0.119)	(0.162)	(0.131)
e_t (EUR/USD)	-2.196	-0.036	-1.258	-0.913
	(1.504)	(0.477)	(1.071)	(0.760)
Constant	-1.405	0.041	0.942	0.911
	(1.692)	(0.278)	(0.672)	(0.566)
TTF total, bp per +10%	8.90^{***}	0.65	-0.80	-0.52
	(3.16)	(0.78)	(1.05)	(0.76)
R^2	0.243	0.173	0.045	0.050
Adjusted R^2	0.233	0.169	0.037	0.042
Robust F (all slopes)	31.4^{***}	17.3^{***}	3.7^{**}	3.8^{**}
Residual s.e. (bp)	28.20	8.00	13.35	10.19
Durbin–Watson	1.85	2.31	1.98	1.87
HAC lags	4	5	5	5
Observations	232	555	362	363
Panel B. Bund yields				
	Bund 2Y	Bund 5Y	Bund 10Y	Bund 30Y
g_t (TTF)	-0.034	-0.045	-0.054	-0.077
	(0.055)	(0.057)	(0.055)	(0.054)
o_t (Brent)	0.297^{***}	0.313^{***}	0.316^{***}	0.336^{***}
	(0.086)	(0.087)	(0.092)	(0.078)
e_t (EUR/USD)	-0.080	-0.314	-0.246	0.016
	(0.464)	(0.610)	(0.640)	(0.602)
Constant	0.447	0.450	0.369	0.284
	(0.346)	(0.349)	(0.364)	(0.389)
TTF total, bp per +10%	-0.32	-0.42	-0.51	-0.73
	(0.52)	(0.54)	(0.52)	(0.52)
R^2	0.030	0.030	0.030	0.034
Adjusted R^2	0.026	0.025	0.025	0.029
Robust F (all slopes)	4.2^{***}	5.3^{***}	4.9^{***}	6.7^{***}
Residual s.e. (bp)	8.84	9.51	9.51	9.71
Durbin–Watson	1.96	2.06	2.04	1.94
HAC lags	5	5	5	5
Observations	620	620	620	620

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 24: OLS transmission regressions, weekly, full sample, part 2: curve slopes

Panel C. Curve slopes				
	2s5s	2s10s	2s30s	5s30s
g_t (TTF)	−0.011	−0.020	−0.043	−0.032
	(0.019)	(0.033)	(0.045)	(0.037)
o_t (Brent)	0.016	0.018	0.039	0.023
	(0.039)	(0.057)	(0.064)	(0.047)
e_t (EUR/USD)	−0.234	−0.166	0.096	0.330
	(0.246)	(0.364)	(0.432)	(0.338)
Constant	0.003	−0.078	−0.163	−0.166
	(0.150)	(0.244)	(0.331)	(0.244)
TTF total, bp per +10%	−0.10	−0.19	−0.41	−0.30
	(0.18)	(0.31)	(0.43)	(0.36)
R^2	0.004	0.001	0.003	0.006
Adjusted R^2	-0.001	-0.003	-0.002	0.001
Robust F (all slopes)	0.5	0.2	0.5	0.7
Residual s.e. (bp)	3.94	6.18	8.34	6.56
Durbin–Watson	2.07	2.04	2.04	2.16
HAC lags	5	5	5	5
Observations	620	620	620	620

Notes: OLS on Friday-to-Friday changes. Weekly TTF and Brent returns are sums of roll-cleaned daily returns; for Bund yields and slopes the market window ends on Thursday so that it closes before the Bundesbank snapshot. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Unit roots and cointegration

Table 25: Unit-root tests on weekly data

Series	Form	ADF p	KPSS p	ZA p	ZA break	Verdict
bund_2y	level	0.974	0.010	0.002	2022-03-04	unit root
bund_2y	first difference	< 0.001	0.082	< 0.001	2023-03-03	stationary
bund_10y	level	0.981	0.010	0.074	2022-03-04	unit root
bund_10y	first difference	< 0.001	0.071	< 0.001	2023-03-03	stationary
bund_30y	level	0.952	0.010	0.460	2022-03-04	unit root
bund_30y	first difference	< 0.001	0.066	< 0.001	2023-10-27	stationary
slope_2s10s	level	0.402	0.010	0.781	2024-07-19	unit root
slope_2s10s	first difference	< 0.001	0.100	< 0.001	2023-06-30	stationary
ln_ttf	level	0.387	0.010	0.307	2021-03-19	unit root
ln_ttf	first difference	< 0.001	0.100	< 0.001	2022-08-26	stationary
ln_brent	level	0.165	0.010	0.258	2020-10-30	unit root
ln_brent	first difference	< 0.001	0.100	< 0.001	2020-04-24	stationary
ils_eur_1y1y	level	0.464	0.010	0.217	2021-07-23	unit root
ils_eur_1y1y	first difference	< 0.001	0.100	< 0.001	2022-06-10	stationary
ois_estr_1y1y	level	0.793	0.010	0.466	2022-03-04	unit root
ois_estr_1y1y	first difference	< 0.001	0.100	< 0.001	2023-03-03	stationary
ois_estr_2y	level	0.780	0.010	0.470	2022-03-04	unit root
ois_estr_2y	first difference	0.066	0.100	0.174	2023-03-03	mixed / borderline
swapsprd_de5y_bp	level	0.557	0.024	0.739	2023-10-20	unit root
swapsprd_de5y_bp	first difference	< 0.001	0.100	< 0.001	2022-10-14	stationary

Notes: ADF and Zivot–Andrews: null of a unit root; KPSS: null of stationarity. KPSS p-values are truncated at 0.01 and 0.10.

Table 26: Cointegration tests on weekly levels

Test	System	Weeks	Statistic	p	Rank (5%)
Engle-Granger	bund_2y $\sim$ ln_ttf	627	-1.17	0.866	0
Engle-Granger	ils_eur_1y1y $\sim$ ln_ttf	562	-2.40	0.323	0
Engle-Granger	ils_eur_1y1y $\sim$ ln_brent	562	-3.31	0.054	0
Engle-Granger	ois_estr_1y1y $\sim$ ils_eur_1y1y	367	-1.35	0.815	0
Engle-Granger	bund_2y $\sim$ ois_estr_2y	368	-1.10	0.883	0
Engle-Granger	bund_2y $\sim$ ils_eur_1y1y	562	-1.93	0.566	0
Johansen trace	bund_2y , ils_eur_1y1y , ln_ttf , ln_brent	562	64.55		3
Johansen trace	bund_2y , ois_estr_1y1y , ils_eur_1y1y , ln_ttf	367	56.58		1

Notes: Engle–Granger: null of no cointegration. Johansen: sequential trace test at 5%.

C Replication

The pipeline is in `src/`: `clean.py` builds the panels, `econometrics.py` runs every test and writes `output/tables/`, `figures.py` draws the charts, `build_workbook.py` assembles the results workbook and `build_paper.py` writes and compiles this paper. Every number in the text is filled in from the result tables at build time.